AYUSH RAJ

B.Tech Computer Science and Engineering
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EDUCATION

B.Tech (Computer Science and Engineering), Birla Institute of Technology, Mesra 2024 – 2028

TECHNICAL SKILLS

Programming & ML: Python, Machine Learning, Deep Learning, Computer Vision (YOLOv8)
ML Systems: Model training & evaluation, real-time inference, data annotation, class imbalance handling
Computer Vision & AI: Object detection, image processing, safety compliance systems (PPE detection)
Explainable AI: SHAP, LIME, model interpretability, error analysis
Tools & Frameworks: PyTorch, OpenCV, Flask, FastAPI, NumPy, Pandas, Git

INDUSTRIAL EXPERIENCE

Machine Learning & Computer Vision Intern Dec 2025 – Jan 2026
@ Bharat Coking Coal Limited (BCCL) Dhanbad, Jharkhand

- Project 1 (Vision system)

  Vision-Based PPE Compliance Detection System (YOLOv8)

  - Engineered a real-time PPE compliance system using YOLOv8 and Flask, trained on 5,000+ annotated images to detect helmets, vests, and masks in industrial environments.
  - Achieved ~77% mAP with <1s inference latency, improving accuracy by 18% over baseline and reducing manual safety monitoring effort by ~80%.
  - Built a predictive analytics platform using FastAPI, Streamlit, and regression models to analyze historical mining data and enhance coal production forecasting and dispatch planning.
  - Integrated an LLM-powered (OpenAI) conversational interface into the dashboard, enabling natural-language querying of predictions and faster decision-making for stakehold.

- Project 2 (Vision system)

  Predictive Analytics Platform for Coal Production & Dispatch (BCCL)

  - Built a multi-output regression system (Random Forest) using Scikit-learn to forecast production, dispatch, and reserves from 3 years of mining data across 7 locations in Dhanbad.
  - Achieved R^2: 0.85–0.92, leveraging feature engineering (lagged production, cost factors, temporal data) and preprocessing pipelines (ColumnTransformer, OneHotEncoder).
  - Developed a FastAPI + Streamlit dashboard with an LLM-powered (OpenAI) interface for natural-language queries on predictions and insights.

PROJECTS

Mental Health Analysis Software 2026
Supervisor: Dr. Alok Mishra, Norwegian University of Science and Technology

- Built a hierarchical mental health classification system using Mental-RoBERTa, separating distress detection from disorder classification.
- Applied class-imbalance mitigation and noise-robust training using weighted loss and label smoothing.
- Performed detailed error analysis using confusion matrices and TP/FP/FN metrics.
- Implemented risk-aware inference logic for safety-critical cases (Depression vs Suicidal) using probability.
- thresholds, achieving >90% system-level accuracy.
- Deployed a modular, production-style ML inference application with persistent model storage and an
- interactive Gradio web interface.

Project: Loan Approval Prediction (ML vs LLM-driven Decision System) 2025

- Built a predictive system comparing ML models (Random Forest, Logistic Regression) with an LLM-based decision engine for loan approvals in the Indian financial context.
- Integrated OpenAI API for contextual risk assessment using applicant data (CIBIL, employment type, regional factors) with explainable AI decisions.
- Developed Python pipelines (Pandas, Scikit-learn) for preprocessing, feature engineering, and model training on 10K+ records(Dataset Source-Credit Risk Dataset), achieving ~89% accuracy.
- Deployed a Flask web app with interactive dashboards; LLM-assisted approach outperformed baseline ML by ~12% in decision quality.

TECHNICAL CO-CURRICULAR

Executive Member @ IEEE Student Branch, BIT Mesra

- Organized and conducted coding contests, technical workshops, and hackathons for students, increasing participation and engagement in the IEEE student community.
- Coordinated with teams to manage event logistics, promotions, and execution, ensuring smooth delivery of technical and co-curricular activities.
- Contributed to fostering a collaborative learning environment by mentoring juniors and promoting technical skill development initiatives.

RESEARCH

Transformer-based Detection of Mental Health Ailments (IEEE Working Paper) 2026

Dr. Kamta Nath Mishra, Birla Institute of Technology Mesra
Dr. Alok Mishra, Norwegian University of Science and Technology

- Proposed a hierarchical formulation for mental health text classification, decomposing the task into distress detection and disorder-specific classification to reduce class entropy and improve robustness on noisy social- media data.
- Leveraged Mental-RoBERTa, a domain-specific pretrained transformer, combined with imbalance-aware training and label smoothing to improve generalization across underrepresented mental health categories.
- Introduced risk-aware post-hoc inference and severity-aware evaluation to address systematic overlap between Depression and Suicidal classes, reducing false negatives in safety-critical scenarios.
- Conducted comprehensive system-level evaluation using accuracy, macro-F1, confusion matrices, and TP/FP/FN analysis, demonstrating >90% screening-level accuracy under clinically aligned metrics.

ACHIEVEMENTS

Bharat Coking Coal Limited (BCCL) | Internship & Project Certification (Dec 2025 – Jan 2026)

- Completed 4-week industrial training on Vision-Based PPE Compliance & ML-driven Safety Systems Internship & Project Certification

COURSES

Stanford OpenCourseWare: CS221 (AI), CS229 (Machine Learning), CS230 (Deep Learning), CS224N (NLP)
Complete Data Science, Machine Learning, Deep Learning & NLP Bootcamp – Krish Naik (Udemy)
Mastering OpenCV: A Practical Guide to Computer Vision (Udemy) Certificate
Hands-On Python Machine Learning with Real World Projects Certificate